Solvency Field Theory
V11 Derivation Companion - Part 2

Galactic Projection, the $4\pi^2$ Spine, the Repaired Lagrangian, Forced Transitions, and the Cosmological Sector

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Companion draft prepared from V11 Master Part 2, the V10.2 projection note, V10.3–V10.9 theorem notes, and the repaired action documents

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Abstract

This companion volume supports Sections 7–11 of the V11 Master Paper. It carries the derivation layer underneath the galactic acceleration scale, the SPARC morphology prediction, the unified S^2 projection family, the $4\pi^2$ structural spine, the repaired five-term Lagrangian, the forced-transition solvency ratio, and the late-universe cosmological sector. The master paper states these pieces as one physical story. This companion separates definitions, assumptions, derivations, branch conditions, and closure status. The dominant equations are intentionally displayed as the main objects of the document:

$$a_{\mathrm{SFT}} = \frac{cH_0}{2\pi}, \qquad \lambda(\alpha) = \sqrt{\pi}\frac{\Gamma((\alpha+4)/2)}{\Gamma((\alpha+5)/2)},$$

$$S_{\mathrm{SFT}} = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \eta\nabla_\mu(\bar{\chi}+\psi)J^\mu_{\mathrm{ops}} + \Lambda_C\mathcal{C}_{\mathrm{min}} - V_{\mathrm{quad}} + \mathcal{L}_m\right],$$

$$V_{\mathrm{quad}} = V_0(e^{-\psi}-1)^2, \qquad \Lambda_X(t) = \frac{D(t)}{C_X(t)}.$$

Where the master gives the working theory, this document gives the mathematical receipt for Part 2.

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1 Role of this companion

This volume mirrors Master Sections 7–11. It does not repeat the scientific-conversational explanation unless that explanation is needed to define a mathematical object. Its function is to make the Part 2 claims auditable.

1.1 Source hierarchy for Part 2

The following hierarchy is used whenever older notes and the V11 Master use different language.

1. **V11 Master Part 2 Final** is the current claim and status source.
2. **SFT Lagrangian, Lagrangian History Main, and Lagrangian History Derivations Companion** are the current action sources. These supersede the earlier V10.7–V10.9 absolute-potential action where they conflict.
3. **V10.2 Projection Geometry Closure** is the derivation source for the unified S^2 projection family, the Gamma-family morphology curve, and the morphology test.
4. **V10.3 Unified Maintenance Closure** is the derivation source for the corrected solvency ratio and the four-regime effective history.
5. **V10.4 Thermal Closure** is the derivation source for the second 2π factor.
6. **V10.5–V10.9** are gate-history sources for Theorem 6a–6c and the pre-repair branch selection sequence.

1.2 What this part proves or audits

Companion Part 2 supports the following master-paper spine:

$$a_{\mathrm{SFT}} = \frac{cH_0}{2\pi}, \tag{1}$$

$$F(x) = 1 - e^{-x}, \qquad x = \frac{g_N}{a_{\mathrm{SFT}}}, \tag{2}$$

$$\nu(y) = \sqrt{1 + \frac{\lambda}{y}}, \qquad g_{\mathrm{eff}} = \nu(y) g_N, \tag{3}$$

$$M(w, P; c) = c\langle w(\theta) P(\theta)\rangle_{S^2}, \tag{4}$$

$$\lambda(\alpha) = \sqrt{\pi}\frac{\Gamma((\alpha+4)/2)}{\Gamma((\alpha+5)/2)}, \tag{5}$$

$$4\pi^2 = (2\pi)_{\mathrm{phase}}(2\pi)_{\mathrm{thermal}}, \tag{6}$$

$$S_{\mathrm{SFT}} = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \eta\nabla_\mu(\bar{\chi} + \psi)J^\mu_{\mathrm{ops}} + \Lambda_C \mathcal{C}_{\mathrm{min}} - V_{\mathrm{quad}} + \mathcal{L}_m\right], \tag{7}$$

$$V_{\mathrm{quad}} = V_0(e^{-\psi} - 1)^2, \tag{8}$$

$$\Lambda_X(t) = \frac{D(t)}{C_X(t)}. \tag{9}$$

1.3 Status language

Status	**Meaning**
Exact	Follows algebraically or variationally inside the stated branch assumptions.
Theorem-level closed	The derivation is carried to the floor inside the stated scaffold.
Structurally closed	The right mathematical object is identified, with implementation details remaining.
Branch-level closed	The result is exact or controlled on the promoted branch, but not yet a fully background-independent or all-branch theorem.
Effective closure	A robust effective model exists, while first-principles histories remain open.
Empirically supported	A data test supports the prediction, but the theorem status is distinct from the empirical status.
Research target	The derivation or empirical program remains explicitly open.

2 Notation and conventions

Planck's constant is written h_{P} in this companion to avoid confusion with the spatial projector $h_{\mu\nu}$:

$$h_{\mathrm{P}} \equiv \text{Planck's constant}, \qquad \hbar = \frac{h_{\mathrm{P}}}{2\pi}. \tag{10}$$

The clock congruence and spatial projector are

$$u^\mu u_\mu = -1, \qquad h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu, \qquad h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu. \tag{11}$$

The covariant operation density, derived in Companion Part 1 and used throughout Part 2, is

$$\rho_{\mathrm{ops}} = \frac{T_{\mu\nu} u^\mu u^\nu + h^{\mu\nu} T_{\mu\nu}}{h_{\mathrm{P}}}. \tag{12}$$

For a perfect fluid in the clock frame,

$$\rho_{\mathrm{ops}} = \frac{\rho + 3p}{h_{\mathrm{P}}}. \tag{13}$$

The reduced Friedmann convention used in several branch equations is

$$\frac{8\pi G}{3} = 1 \quad \text{or, when energy density rather than mass density is explicit,} \quad \frac{8\pi G}{3c^2} = 1. \tag{14}$$

When a formula is quoted in reduced units, the companion states so explicitly.

3 Claim ledger for Master Sections 7–11

Master	Load-bearing claim	Companion proof block	Status
7.1	$a_{\mathrm{SFT}} = cH_0/(2\pi)$ is derived, not fitted.	Sec. 4	Derived scale
7.2	The completion law is exponential because forwarding fractions compose multiplicatively.	Sec. 5	Structurally closed
7.2	$g_{\mathrm{eff}} = g_N\sqrt{1+\lambda/y}$ is the weak-field response convention.	Sec. 6	Branch response
7.4	Wall, filament, sphere, and disk coefficients are one S^2 projection family.	Sec. 7	Theorem-level closed
7.4–7.5	The Gamma-family morphology curve predicts the sign and endpoint ordering of λ_{obs}.	Sec. 8	v1 empirical closure
8	$4\pi^2$ decomposes as phase 2π times thermal 2π.	Sec. 9	Theorem-level closed
8.3	$\epsilon_\Phi = 1/(4\pi^2)$ follows from an Onsager interface law at transport level.	Sec. 10	Transport-level derived
9	The repaired action is a five-term effective covariant action for local lag dynamics about the cosmological branch.	Sec. 11	Live action candidate
9.4	The absolute potential fails exact static vacuum; the quadratic relative potential repairs it.	Sec. 12	Exact obstruction/repair
9.5	The repaired action has exact exterior branch collapse and a branch-level modified TOV interior.	Sec. 13	Exact exterior; branch interior
10	Shatter and engine crash are two crossings of one solvency ratio.	Sec. 14	Effective closure
11	The cosmic-web ladder and observer shell correction build the late-universe $w(z)$ pipeline.	Sec. 15	Pipeline-level reconstruction

4 Galactic acceleration scale

Master claim. The SFT galactic acceleration scale is $a_{\mathrm{SFT}} = cH_0/(2\pi)$, derived from the phase/cycle leg of the $4\pi^2$ spine rather than fitted to rotation curves.

The dimensionally available acceleration constructed from the Hubble rate and the speed of light is

$$a_H = cH_0. \tag{15}$$

SFT does not use angular phase as its primitive count unit. It counts complete phase cycles. The conversion between angular and cycle variables contributes

$$h = 2\pi\hbar. \tag{16}$$

The galactic threshold is therefore the cycle-normalized Hubble acceleration:

$$\boxed{a_{\mathrm{SFT}} = \frac{cH_0}{2\pi}.} \tag{17}$$

Using the H_0 value carried in the SFT empirical pipeline gives

$$a_{\mathrm{SFT}} \simeq 1.04 \times 10^{-10}\ \mathrm{m\,s^{-2}}. \tag{18}$$

The empirical MOND scale is approximately

$$a_0 \simeq 1.2 \times 10^{-10}\ \mathrm{m\,s^{-2}}, \tag{19}$$

so the residual is

$$\frac{a_0 - a_{\mathrm{SFT}}}{a_0} \simeq 0.13. \tag{20}$$

Status. The scale a_{SFT} is derived from cH_0 and the cycle/phase factor. The 13% offset relative to empirical a_0 remains Research Target 3.

5 Multiplicative forwarding and the completion law

Master claim. The local completion law $F(x) = 1 - e^{-x}$ follows from multiplicative forwarding. Local structures absorb part of the maintenance burden; the unabsorbed fraction forwards outward.

Let $U(x)$ denote the unabsorbed or forwarded fraction after passing through a structure of dimensionless screening depth x. If two independent layers are traversed in sequence, the forwarded fractions multiply:

$$U(x_1 + x_2) = U(x_1)U(x_2). \tag{21}$$

Assuming continuity and normalization $U(0) = 1$, the only nontrivial solution is exponential. To see this, set $x_2 = \delta x$ and expand:

$$U(x + \delta x) = U(x)U(\delta x). \tag{22}$$

With

$$U(\delta x) = 1 - k\delta x + O(\delta x^2), \tag{23}$$

one obtains

$$\frac{dU}{dx} = -kU, \tag{24}$$

and hence

$$U(x) = e^{-kx}. \tag{25}$$

The completion fraction is

$$F(x) = 1 - U(x) = 1 - e^{-kx}. \tag{26}$$

The V10.5 bridge analysis selects the transmission-side coefficient $k = 1$, giving

$$\boxed{F(x) = 1 - e^{-x}, \qquad x = \frac{g_N}{a_{\mathrm{SFT}}}.} \tag{27}$$

Status. The exponential form is structurally closed from multiplicative composition. The selection $k = 1$ is the V10.5 transmission-side branch choice; environment-dressed effective depth remains an implementation target.

6 Weak-field interpolating response

Master claim. The SFT interpolating function is used through $g_{\text{eff}} = \nu(y) g_N$ with $\nu(y) = \sqrt{1 + \lambda/y}$ and $y = g_N/a_{\text{SFT}}$.

The weak-field SFT response is written in quadratic acceleration form:

$$g_{\text{eff}}^2 = g_N^2 + \lambda a_{\text{SFT}} g_N. \tag{28}$$

Dividing by g_N^2 gives

$$\left(\frac{g_{\text{eff}}}{g_N}\right)^2 = 1 + \lambda \frac{a_{\text{SFT}}}{g_N}. \tag{29}$$

Define

$$y \equiv \frac{g_N}{a_{\text{SFT}}}. \tag{30}$$

Then

$$\boxed{\nu(y) \equiv \frac{g_{\text{eff}}}{g_N} = \sqrt{1 + \frac{\lambda}{y}}, \qquad g_{\text{eff}} = \nu(y) g_N.} \tag{31}$$

In the high-acceleration regime $y \gg 1$,

$$\nu(y) = 1 + \frac{\lambda}{2y} + O(y^{-2}), \tag{32}$$

so Newtonian/GR behavior is recovered. In the low-acceleration regime $y \ll 1$,

$$g_{\text{eff}} \sim g_N \sqrt{\frac{\lambda}{y}} = \sqrt{\lambda a_{\text{SFT}} g_N}, \tag{33}$$

which has the MOND-like square-root scaling.

Status. The weak-field interpolation convention is a branch response law tied to the screen-projection coefficient λ. The acceleration scale is fixed; morphology enters through λ.

7 Unified projection family on S^2

Master claim. The cosmological softening coefficients and galactic response coefficients are members of one spherical projection family.

Define the normalized spherical average for an axisymmetric function $Q(\theta)$ by

$$\langle Q\rangle_{S^2} = \frac{1}{2}\int_0^\pi Q(\theta)\sin\theta\, d\theta. \tag{34}$$

The unified family is

$$\boxed{M(w,P;c) = c\langle w(\theta)P(\theta)\rangle_{S^2}.} \tag{35}$$

Here $w(\theta)$ is the participation weight, $P(\theta)$ is the projection kernel, and c is the sector prefactor.

7.1 Backbone moments

Define

$$J_n \equiv \langle \sin^n\theta\rangle_{S^2} = \frac{1}{2}\int_0^\pi \sin^{n+1}\theta\, d\theta. \tag{36}$$

Using the standard integral

$$\int_0^\pi \sin^m\theta\, d\theta = \sqrt{\pi}\frac{\Gamma((m+1)/2)}{\Gamma((m+2)/2)}, \tag{37}$$

we obtain

$$\boxed{J_n = \frac{\sqrt{\pi}}{2}\frac{\Gamma((n+2)/2)}{\Gamma((n+3)/2)}.} \tag{38}$$

The needed moments are

$$J_0 = 1, \qquad J_2 = \frac{2}{3}, \qquad J_3 = \frac{3\pi}{16}. \tag{39}$$

Since $\cos^2\theta = 1-\sin^2\theta$,

$$\langle\cos^2\theta\rangle_{S^2} = 1 - J_2 = \frac{1}{3}. \tag{40}$$

7.2 The four coefficients

The cosmological coefficients use $c = 1$:

$$s_{\text{wall}} = \langle\cos^2\theta\rangle_{S^2} = \frac{1}{3}, \tag{41}$$

$$s_{\text{filament}} = \langle\sin^2\theta\rangle_{S^2} = \frac{2}{3}. \tag{42}$$

The galactic coefficients use the quadratic weak-field prefactor $c = 2$:

$$\lambda_{\text{sph}} = 2\langle\sin^2\theta\rangle_{S^2} = 2J_2 = \frac{4}{3}, \tag{43}$$

$$\lambda_{\text{disk}} = 2\langle\sin\theta\sin^2\theta\rangle_{S^2} = 2J_3 = \frac{3\pi}{8}. \tag{44}$$

Thus

$$\boxed{s_{\text{wall}} = \frac{1}{3}, \qquad s_{\text{filament}} = \frac{2}{3}, \qquad \lambda_{\text{sph}} = \frac{4}{3}, \qquad \lambda_{\text{disk}} = \frac{3\pi}{8}.} \tag{45}$$

7.3 Structural identities

The first identity is completeness:

$$s_{\text{wall}} + s_{\text{filament}} = \frac{1}{3} + \frac{2}{3} = 1. \tag{46}$$

The second identity is the scale bridge:

$$\lambda_{\text{sph}} = 2s_{\text{filament}}. \tag{47}$$

The third identity is the morphology bridge: the wall and disk coefficients live in the same S^2 kernel family, but require a continuous participation interpolation rather than a one-step algebraic identity.

Status. The projection unification is theorem-level closed inside the stated S^2 construction. The sector prefactors differ because the cosmological burden correction is linear while the galactic weak-field response is quadratic.

8 Gamma-family morphology curve

Master claim. The morphology coefficient λ is not universal. It follows a continuous projection curve connecting the spherical and disk endpoints.

Use the participation family

$$w(\theta;\alpha) = \sin^{\alpha}\theta, \qquad 0 \le \alpha \le 1. \tag{48}$$

Here $\alpha = 0$ is the isotropic spherical endpoint and $\alpha = 1$ is the ideal thin-disk endpoint in the V10.2 derivation convention. The response coefficient is

$$\lambda(\alpha) = 2\langle \sin^{\alpha}\theta \sin^2\theta \rangle_{S^2} \tag{49}$$

$$= 2J_{\alpha+2}. \tag{50}$$

Using the moment formula gives

$$\boxed{\lambda(\alpha) = \sqrt{\pi}\frac{\Gamma((\alpha+4)/2)}{\Gamma((\alpha+5)/2)}, \qquad 0 \le \alpha \le 1.} \tag{51}$$

The endpoints are

$$\lambda(0) = \sqrt{\pi}\frac{\Gamma(2)}{\Gamma(5/2)} = \frac{4}{3}, \tag{52}$$

$$\lambda(1) = \sqrt{\pi}\frac{\Gamma(5/2)}{\Gamma(3)} = \frac{3\pi}{8}. \tag{53}$$

Thus the curve interpolates from sphere to disk:

$$\boxed{\lambda_{\text{sph}} = \lambda(0) = \frac{4}{3}, \qquad \lambda_{\text{disk}} = \lambda(1) = \frac{3\pi}{8}.} \tag{54}$$

Remark. The derivation convention uses α as the exponent in $w(\theta;\alpha) = \sin^{\alpha}\theta$, with $0 \le \alpha \le 1$. If the master text uses α more generically as a flattening coordinate, the companion should be read as fixing the mathematical convention: any observational morphology coordinate must map monotonically onto this exponent or an equivalent reparameterization of it.

8.1 Empirical morphology test

The empirical test proceeds by profiling per-galaxy λ_{obs} over stellar mass-to-light nuisance grids, then testing whether λ_{obs} orders with morphology. In the V10.2 interior SPARC sample,

$$n = 131. \tag{55}$$

The primary rank correlation is

$$\boxed{\rho(T, \lambda_{\text{obs}}) = -0.3005, \qquad p = 4.87 \times 10^{-4}.} \tag{56}$$

The inclination-controlled result is

$$\boxed{\rho(T, \lambda_{\text{obs}} \mid i) = -0.2732, \qquad p = 0.0016.} \tag{57}$$

The early-type endpoint lands at

$$\lambda_{\text{early}} = 1.340 \tag{58}$$

against the spherical target

$$\lambda_{\text{sph}} = \frac{4}{3} = 1.333\ldots. \tag{59}$$

Status. The morphology theorem is at v1 empirical closure: the derived curve exists, the sign is correct, the Hubble-type correlation is significant, and the endpoint lands near the spherical target. The direct photometric map from structural observables to the projection exponent remains open.

9 The $4\pi^2$ structural identity

Master claim. The factor $4\pi^2$ is not a single unexplained constant. It decomposes into one phase-counting factor and one horizon-thermal factor.

The decomposition is

$$\boxed{4\pi^2 = (2\pi)_{\text{phase}}(2\pi)_{\text{thermal}}.} \tag{60}$$

9.1 The phase factor

SFT counts maintenance in complete cycles. Standard angular quantum notation counts in radians:

$$E = \hbar\omega = h_{\text{P}}\nu, \qquad \omega = 2\pi\nu, \qquad h_{\text{P}} = 2\pi\hbar. \tag{61}$$

Any interface between complete-cycle SFT counting and angular QFT notation carries the conversion

$$(2\pi)_{\text{phase}} = \frac{h_{\text{P}}}{\hbar}. \tag{62}$$

9.2 The thermal factor

The second factor enters through horizon thermality. The Unruh temperature for acceleration a is

$$T_U = \frac{\hbar a}{2\pi c k_B}. \tag{63}$$

The 2π is not a mere convention. It reflects KMS periodicity in imaginary Rindler time:

$$\tau_E \sim \tau_E + \frac{2\pi c}{a}. \tag{64}$$

In Jacobson-style horizon thermodynamics, the Einstein coupling inherits this horizon-thermal conversion. SFT's $4\pi^2$ spine appears where phase accounting meets gravitational response.

9.3 Appearance ledger

#	Context	Where the factor enters	Status
1	Modified Friedmann	$H^2_{\text{SFT}}/H^2_{\text{std}} = 1/(4\pi^2)$	Derived
2	Galactic scale	$a_{\text{SFT}} = cH_0/(2\pi)$ carries the phase/cycle leg	Derived scale
3	Count equation	Complete Compton cycles to Planck-area capacity	Derived
4	Projection geometry	Same S^2 angular family generates cosmological softening and galactic response coefficients	Theorem closed
5	CMB transfer	$\epsilon_\Phi = 1/(4\pi^2)$ at the relic-photon interface	Onsager-derived

Status. Theorem 2 is closed at the current scaffold level for the first four appearances. Appearance 5 is Onsager-derived at the transport level; its exact interpretive relation to the phase-times-thermal decomposition remains part of the Theorem 3 frontier.

10 Onsager closure of the CMB transfer coefficient

Master claim. The CMB transfer coefficient $\epsilon_\Phi = 1/(4\pi^2)$ is derived from a minimal near-equilibrium Onsager interface law at transport level.

At the shatter-radiation interface, a relic phase perturbation must be converted into a photon-fluid response. The interface is dissipative; therefore the minimal near-equilibrium form is an Onsager relation:

$$J_a = \sum_b L_{ab} X_b, \tag{65}$$

where J_a are fluxes, X_b are thermodynamic forces, and the dissipation functional is quadratic:

$$\mathcal{D} = \frac{1}{2} \sum_{ab} L_{ab} X_a X_b. \tag{66}$$

The SFT transfer block separates into a transport coefficient and unavoidable interface Jacobians:

$$\epsilon_\Phi = J_{\text{phase}} J_{\text{interface}} R_\Phi. \tag{67}$$

The minimal closure isolates the bookkeeping Jacobians:

$$J_{\text{phase}} = \frac{1}{2\pi}, \qquad J_{\text{interface}} = \frac{1}{2\pi}. \tag{68}$$

Thus the bookkeeping/interface part is

$$\boxed{\epsilon_\Phi = \frac{1}{4\pi^2}} \tag{69}$$

when the remaining transport block is normalized in the minimal Onsager variables.

Status. The coefficient value is transport-level derived. The still-open issue is not the numerical value, but the interpretation: whether the interface Jacobians are the same phase-times-thermal pair as the first four appearances or a consistent extension of the same spine.

11 The repaired five-term Lagrangian

Master claim. The live action is a repaired effective covariant action for local lag dynamics about the cosmological bookkeeping branch. It contains no kinetic term for the bookkeeping scalar and no freely propagating dark-sector field.

The live repaired action is

$$\boxed{S_{\mathrm{SFT}} = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi G} - \eta\nabla_\mu(\bar{\chi}+\psi)J^\mu_{\mathrm{ops}} + \Lambda_C\mathcal{C}_{\mathrm{min}} - V_{\mathrm{quad}} + \mathcal{L}_m\right].} \tag{70}$$

The bookkeeping scalar is decomposed as

$$\chi(x) = \bar{\chi}(x) + \psi(x), \tag{71}$$

where $\bar{\chi}$ is the cosmological/background clock branch and ψ is the local lag departure from that branch.

The clock congruence is

$$\boxed{u^\mu = \frac{\nabla^\mu(\bar{\chi}+\psi)}{\sqrt{-\nabla_\alpha(\bar{\chi}+\psi)\nabla^\alpha(\bar{\chi}+\psi)}}.} \tag{72}$$

The operation current is

$$J^\mu_{\mathrm{ops}} = \rho_{\mathrm{ops}}u^\mu. \tag{73}$$

The count sector is

$$\boxed{\mathcal{C}_{\mathrm{min}} = H_\chi^2 - \alpha(E_\chi + V_{\mathrm{quad}}),} \tag{74}$$

with

$$H_\chi = \frac{1}{3}\nabla_\mu u^\mu, \qquad E_\chi = T_{\mu\nu}u^\mu u^\nu, \qquad \alpha = \frac{8\pi G}{3}\beta, \qquad \beta = \frac{1}{4\pi^2}. \tag{75}$$

The repaired potential is

$$\boxed{V_{\mathrm{quad}} = V_0(e^{-\psi} - 1)^2.} \tag{76}$$

Status. This is the current live repaired action candidate. It is effective and background-relative: final background-independent packaging of $\bar{\chi}$ remains open.

12 Source-sector variation and the driver-not-tank result

Master claim. The source coupling behaves as a driver of the bookkeeping variable, not as a stored energy tank in the FLRW Hamiltonian. Its hidden χ-dependence is controlled by clock-frame momentum flux.

Let

$$X \equiv -\nabla_\mu \chi \nabla^\mu \chi > 0, \qquad s \equiv \sqrt{X}, \qquad u^\mu = \frac{\nabla^\mu \chi}{s}. \tag{77}$$

Holding the metric fixed during the χ-variation,

$$\delta X = -2\nabla_\mu \chi \nabla^\mu \delta\chi, \qquad \delta s = -u^\mu \nabla_\mu \delta\chi. \tag{78}$$

The key identity is

$$\boxed{\delta u_\mu = \frac{1}{s} h_\mu{}^\nu \nabla_\nu \delta\chi.} \tag{79}$$

Thus δu_μ is purely spatial in the clock frame.

Define the clock-frame momentum density

$$q^\mu \equiv -h^\mu{}_\alpha T^{\alpha\beta} u_\beta, \qquad q^\mu u_\mu = 0. \tag{80}$$

The hidden variation of the operation density collapses to

$$\boxed{\delta\rho_{\rm ops} = -\frac{4}{h_{\rm P} s} q^\mu \nabla_\mu \delta\chi.} \tag{81}$$

The source Lagrangian is

$$\mathcal{L}_{\rm src} = -\eta \nabla_\mu \chi J^\mu_{\rm ops}. \tag{82}$$

After integrating by parts, the source contribution to the χ equation is

$$\boxed{E^{\rm src}_\chi = \eta \left[\nabla_\mu J^\mu_{\rm ops} + \frac{4}{h_{\rm P}} \nabla_\mu q^\mu \right].} \tag{83}$$

In homogeneous FLRW with a comoving perfect fluid,

$$q^\mu = 0. \tag{84}$$

The source term reduces to a momentum shift. With lapse $N(t)$ and homogeneous $\chi(t)$,

$$\mathcal{L}^{\rm FLRW}_{\rm src} = -\eta a^3 \dot\chi \frac{\rho + 3p}{h_{\rm P}}. \tag{85}$$

The lapse cancels:

$$\boxed{\frac{\partial \mathcal{L}^{\rm FLRW}_{\rm src}}{\partial N} = 0.} \tag{86}$$

The conjugate momentum is shifted:

$$\pi_\chi = a^3 \left(\frac{K}{N} \dot\chi - \eta \frac{\rho + 3p}{h_{\rm P}} \right). \tag{87}$$

In the repaired action, the kinetic term is removed, so the surviving branch is constrained rather than freely propagating.

Status. The hidden source variation is controlled by clock-frame momentum flux. In comoving FLRW sectors, the source is a driver, not a Hamiltonian energy tank.

13 Count sector and minisuperspace branches

Master claim. The minimal count sector carries the $4\pi^2$ normalization in the dust branch and preserves the promoted radiation branch.

The minimal count constraint is

$$\mathcal{C}_{\min} = H_\chi^2 - \alpha(E_\chi + V_{\text{quad}}). \tag{88}$$

In homogeneous FLRW,

$$H_\chi = H, \qquad E_\chi = \rho. \tag{89}$$

Thus

$$\mathcal{C}_{\min} = H^2 - \alpha(\rho + V_{\text{quad}}). \tag{90}$$

The constraint $\mathcal{C}_{\min} = 0$ gives

$$H^2 = \alpha(\rho + V_{\text{quad}}). \tag{91}$$

With

$$\alpha = \frac{8\pi G}{3}\frac{1}{4\pi^2}, \tag{92}$$

one obtains the SFT normalization.

13.1 Dust branch

For dust,

$$w = 0, \qquad p = 0, \tag{93}$$

and the source driving vanishes in the slaved background branch. On $\psi = 0$,

$$V_{\text{quad}} = 0. \tag{94}$$

Therefore

$$\boxed{H^2 = \frac{8\pi G}{3}\frac{1}{4\pi^2}\rho.} \tag{95}$$

In reduced units,

$$\boxed{H^2 = \frac{\rho}{4\pi^2}.} \tag{96}$$

13.2 Radiation branch

For radiation,

$$w = \frac{1}{3}, \qquad p = \frac{\rho}{3}. \tag{97}$$

The source driving is active. In the slaved/effective branch inherited from V10.7–V10.8, the background equation is

$$H^2 - \beta A\rho H - \beta\rho = 0, \qquad \beta = \frac{1}{4\pi^2}, \tag{98}$$

with $A = 2$ for the normalized radiation source. The positive-root solution gives

$$\frac{H^2}{\rho} = \frac{1}{4}\left(\beta A\sqrt{\rho} + \sqrt{\beta^2 A^2 \rho + 4\beta}\right)^2. \tag{99}$$

As $\rho \to 0$,

$$\boxed{\lim_{\rho\to 0} \frac{H^2}{\rho} = \beta = \frac{1}{4\pi^2}.} \tag{100}$$

Status. Dust is exact on the promoted branch. Radiation is asymptotically exact, with finite-density corrections carried by the slaved source branch.

14 Potential repair and exact static vacuum

Master claim. The original absolute exponential potential obstructs exact static vacuum. The repaired quadratic background-relative potential is the first candidate satisfying zero value, zero force, and positive restoring curvature on the vacuum branch.

The pre-repair potential was

$$V_{\text{abs}} = V_0 e^{-\chi}. \tag{101}$$

On a static Killing-aligned clock branch,

$$\chi = \chi_0 + \kappa t. \tag{102}$$

In static vacuum,

$$T_{\mu\nu} = 0, \qquad J^{\mu}_{\text{ops}} = 0, \qquad E_{\chi} = 0, \qquad H_{\chi} = 0. \tag{103}$$

However

$$V_{\text{abs}} = V_0 e^{-\chi_0 - \kappa t} \neq 0, \tag{104}$$

and the count sector becomes

$$\mathcal{C}^{\text{vac}}_{\text{min}} = -\alpha V_{\text{abs}}. \tag{105}$$

The absolute potential therefore leaves a residual non-Einstein vacuum load.

The repair sequence is

$$V_0 e^{-\chi} \longrightarrow V_0 e^{-(\chi - \bar{\chi})} \tag{106}$$

$$\longrightarrow V_0 (e^{-(\chi - \bar{\chi})} - 1) \tag{107}$$

$$\longrightarrow V_0 (e^{-(\chi - \bar{\chi})} - 1)^2. \tag{108}$$

With

$$\psi = \chi - \bar{\chi}, \tag{109}$$

the repaired potential is

$$\boxed{V_{\text{quad}} = V_0 (e^{-\psi} - 1)^2.} \tag{110}$$

On $\psi = 0$,

$$V_{\text{quad}}\Big|_{\psi=0} = 0, \tag{111}$$

$$\frac{dV_{\text{quad}}}{d\psi}\Big|_{\psi=0} = 0, \tag{112}$$

$$\frac{d^2 V_{\text{quad}}}{d\psi^2}\Big|_{\psi=0} = 2V_0 > 0. \tag{113}$$

Result. The repaired potential removes both the metric-side vacuum load and the χ-equation-side force on the background branch while retaining positive restoring curvature.

15 Action outputs: exterior, cosmology, and interior

15.1 Exact exterior collapse

On the background-matched static vacuum branch,

$$\psi = 0, \qquad T_{\mu\nu} = 0, \qquad H_\chi = 0. \tag{114}$$

Then

$$V_{\text{quad}} = 0, \qquad \frac{dV_{\text{quad}}}{d\psi} = 0, \qquad J^\mu_{\text{ops}} = 0, \qquad E_\chi = 0, \qquad \mathcal{C}_{\min} = 0. \tag{115}$$

Every non-Einstein sector vanishes and the action reduces to

$$\boxed{S_{\text{vac}} = \int d^4x\sqrt{-g}\frac{R}{16\pi G}.} \tag{116}$$

Thus the vacuum field equation is

$$G_{\mu\nu} = 0. \tag{117}$$

For a static, spherically symmetric, asymptotically flat exterior, Schwarzschild follows exactly.

15.2 Modified interior branch

For a static spherical metric

$$ds^2 = -e^{2\Phi(r)}dt^2 + \frac{dr^2}{A(r)} + r^2 d\Omega^2, \qquad A(r) = 1 - \frac{2m(r)}{r}, \tag{118}$$

and perfect fluid

$$T^\mu{}_\nu = \text{diag}(-\rho, p, p, p), \tag{119}$$

the repaired $\psi = 0$ branch gives the exact mass equation

$$\boxed{m'(r) = 4\pi r^2(1 + \Lambda_C\alpha)\rho(r).} \tag{120}$$

The radial branch stress from the source sector is

$$T^t{}_t(\text{src}) = 0, \qquad T^r{}_r(\text{src}) = -\frac{\eta\kappa}{h_{\text{P}}}e^{-\Phi(r)}(\rho + 3p). \tag{121}$$

Using perfect-fluid closure for the matter+count part gives

$$p^{(r)}_{\text{eff}} = (1 + \Lambda_C\alpha)p - \frac{\eta\kappa}{h_{\text{P}}}e^{-\Phi}(\rho + 3p). \tag{122}$$

Therefore

$$\boxed{\Phi'(r) = \frac{m(r) + 4\pi r^3\left[(1 + \Lambda_C\alpha)p - \frac{\eta\kappa}{h_{\text{P}}}e^{-\Phi}(\rho + 3p)\right]}{r[r - 2m(r)]}.} \tag{123}$$

With separate matter conservation on the effective-fluid branch,

$$\boxed{p'(r) = -(\rho + p)\Phi'(r).} \tag{124}$$

Status. Exterior Schwarzschild recovery is exact on the background-matched static vacuum branch. The interior mass equation is exact on the repaired $\psi = 0$ branch; the radial pressure/TOV system is branch-level/effective-fluid closure pending full radial variational closure.

16 Forced thermodynamic transitions

Master claim. The shatter and the engine crash are two channel-critical crossings of one solvency ratio.

Define the maintenance demand

$$D(t) = H(t)I(t), \tag{125}$$

where $I(t)$ is the relevant cosmological ledger content/capacity measure. Let $C_X(t)$ be the capacity of channel X to service the demand. The channel solvency ratio is

$$\boxed{\Lambda_X(t) = \frac{D(t)}{C_X(t)} = \frac{H(t)I(t)}{C_X(t)}.} \tag{126}$$

The orientation is

$$\Lambda_X < 1 \quad \text{solvent}, \qquad \Lambda_X = 1 \quad \text{critical}, \qquad \Lambda_X > 1 \quad \text{insolvent}. \tag{127}$$

A forced transition occurs when

$$\Lambda_X(t) = 1 \tag{128}$$

and the dominant channel can no longer carry the maintenance demand.

The four-regime history is

$$\boxed{\text{shared} \longrightarrow \text{independent} \longrightarrow \text{active} \longrightarrow \text{passive}.} \tag{129}$$

The shatter is the shared-to-independent transition. The engine crash is the active-to-passive transition. The independent bridge channel prevents a post-shatter insolvent gap.

16.1 Robustness basin

The effective history occupies a finite basin. The pinned sweep yielded

$$\frac{489}{972} = 0.503 \tag{130}$$

accepted models. The local-release sweep yielded

$$\frac{1836}{8748} = 0.210 \tag{131}$$

accepted models. The sensitivity ranking identifies the shared-channel weakening as the most fragile direction and the passive channel as the most tolerant.

Status. Theorem 1 is at effective closure: the corrected ratio, four-regime history, and robustness basin are established, while first-principles channel histories remain open.

17 Late-universe cosmological sector

Master claim. The dark-energy equation-of-state reconstruction is built from the four-state cosmic-web ladder, unified projection corrections, observer-side shell correction, and channel-history weights.

17.1 Four-state ladder

The cosmic web is divided into four environmental states:

$$\text{void}, \qquad \text{wall}, \qquad \text{filament}, \qquad \text{halo}. \tag{132}$$

The ladder weights are derived from axis factorization:

1. each spatial axis contributes independently,
2. each axis is binary, collapsed or uncollapsed,
3. full collapse gives maximum local absorption.

The geometric projection corrections are then supplied by the same S^2 family used in the galactic sector.

17.2 Observer-side shell correction

The multiplicative forwarding law along a line of sight gives

$$U_{\text{obs}}(\Omega) = U_{\text{global}} \prod_i U_i(\Omega). \tag{133}$$

Taking logarithms,

$$\ln U_{\text{obs}}(\Omega) = \ln U_{\text{global}} + \sum_i \ln U_i(\Omega). \tag{134}$$

Sky-averaging and linearizing in the weak/moderate screening regime yields the observer-side correction

$$\boxed{\Delta U_{\text{rec}}(z) = -\frac{1}{\chi(z)} \sum_i f_{\text{cov},i}\,[U_i - U_{\text{global}}(z)] L_i(z).} \tag{135}$$

Here $f_{\text{cov},i}$ is covering fraction, U_i is the structure's screening level, $U_{\text{global}}(z)$ is the global background burden, and $L_i(z)$ is path length.

17.3 Equation-of-state reconstruction

The SFT late-universe pipeline combines

$$\text{four-state ladder} + extprojectioncorrections + extchannelweights + \text{observer correction} \tag{136}$$

to reconstruct an effective $w(z)$. The master status is pipeline-level reconstruction agreement with DESI DR2 features, including a crossing near

$$w = -1, \qquad z \simeq 0.3\text{–}0.5. \tag{137}$$

The CPL coefficient reconciliation under the corrected chain remains the highest-priority audit item.

Status. The cosmological sector is structurally integrated with the projection and solvency-transition framework. The DESI comparison is pipeline-level reconstruction agreement; CPL reconciliation and cross-simulation validation remain research targets.

18 Part 2 audit table

Object	Derived or stated result	Closure status
Acceleration scale	$a_{\mathrm{SFT}} = cH_0/(2\pi)$	Derived scale; 13% residual open
Completion law	$F(x) = 1 - e^{-x}$	Structurally closed; environment dressing open
Weak-field interpolation	$g_{\mathrm{eff}} = g_N\sqrt{1+\lambda/y}$	Branch response convention
Projection family	$M(w, P; c) = c\langle wP\rangle_{S^2}$	Theorem-level closed
Morphology curve	$\lambda(\alpha) = \sqrt{\pi}\Gamma((\alpha+4)/2)/\Gamma((\alpha+5)/2)$	v1 empirical closure; photometric map open
$4\pi^2$ spine	$(2\pi)_{\mathrm{phase}}(2\pi)_{\mathrm{thermal}}$	Closed at theorem level for main spine
CMB transfer coefficient	$\epsilon_\Phi = 1/(4\pi^2)$	Onsager-derived at transport level
Action	Five-term repaired effective action	Live repaired candidate
Potential repair	$V_0(e^{-\psi} - 1)^2$	Exact repair of static-vacuum obstruction
Exterior branch	$S_{\mathrm{vac}} = \int \sqrt{-g}R/(16\pi G)$	Exact on background-matched branch
Interior branch	Modified TOV skeleton	Branch-level/effective-fluid closure
Solvency ratio	$\Lambda_X = D/C_X$	Effective closure with robustness basin
Cosmological sector	Four-state ladder + observer correction + DESI pipeline	Pipeline-level reconstruction; audit targets remain

19 Open items carried forward

The following open items are intentionally not closed in this companion part:

1. A direct structural-photometry map to the projection exponent α.
2. The 13% residual between a_{SFT} and empirical a_0.
3. The final interpretation of the CMB transfer $4\pi^2$ relative to the phase-times-thermal spine.
4. Background-independent packaging of $\bar{\chi}$.
5. Full radial variational closure of the modified TOV branch.
6. First-principles derivation of the four channel histories.
7. CPL coefficient reconciliation and cross-simulation validation of the cosmological sector.

Status. The purpose of Part 2 is not to hide these openings but to locate them precisely. The established structure is the projection/action/transition spine; the remaining work is refinement, background packaging, and empirical/variational completion.